STS- & TTT- trained STT & TTS model

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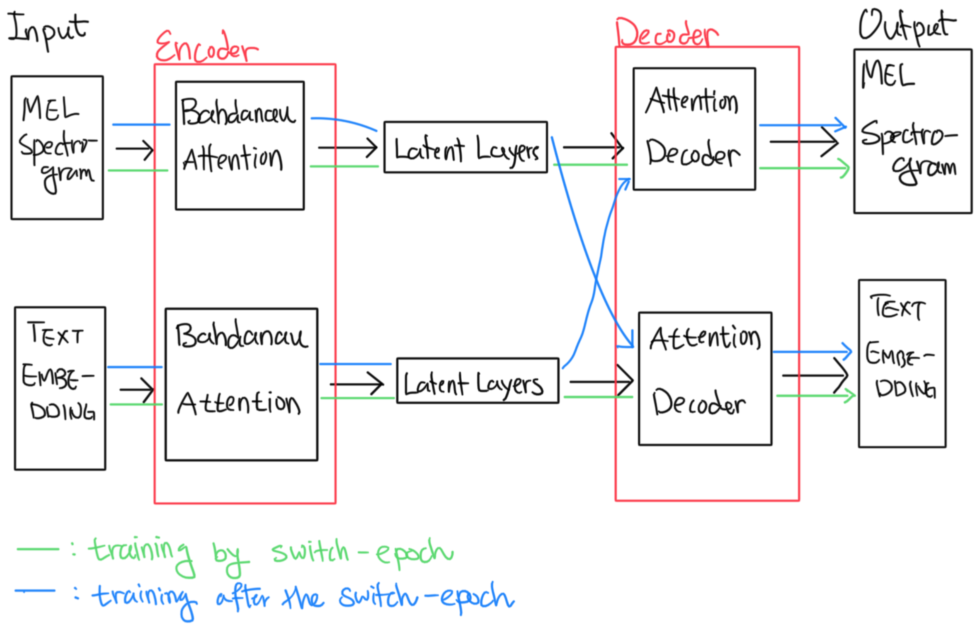
Old failed model



- Train the STS, TTT autoencoders separately up to the **switch epoch**
- Train the STT and TTS models after the switch epoch
  - Only update the latent layers at this stage
- Also tried with transformer encoders & decoders

The problem with the old model

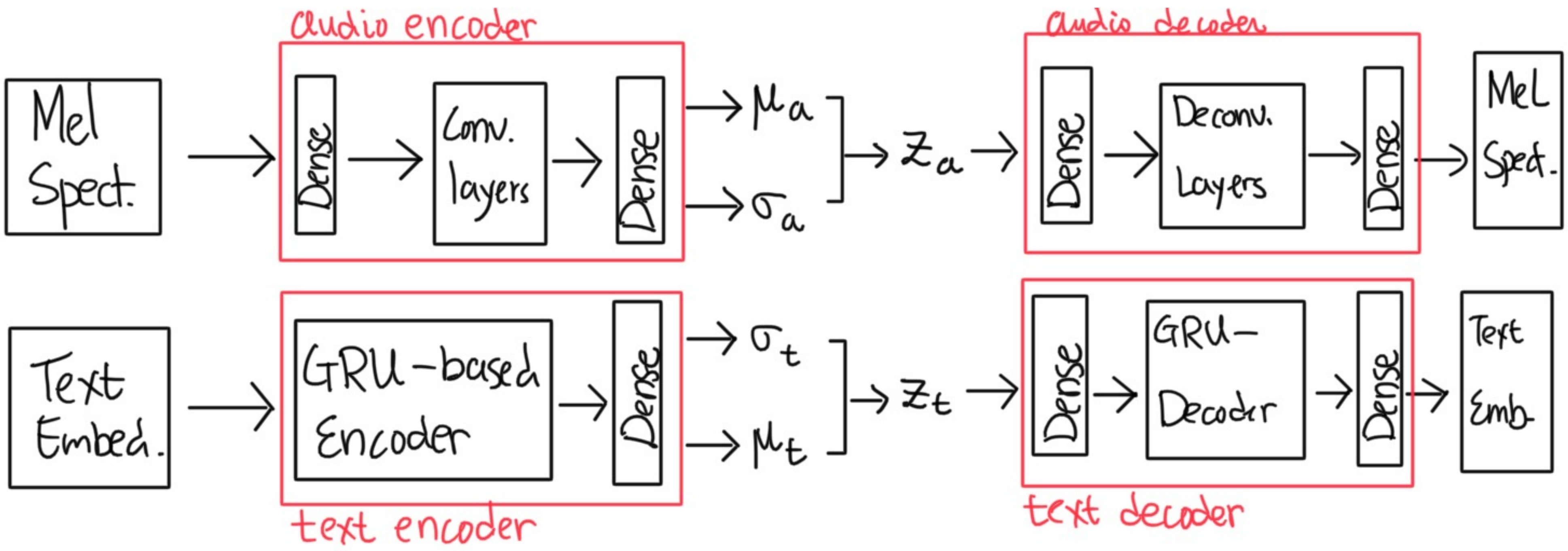
- The intuition:
  - STS model's decoder/encoder (de-)compresses the audio data
  - TTT model's decoder/encoder (de-)compresses the text data
- Natural question is: "can we swap the encoders to produce the STT and TTS models?"

BUT…

Problems…

- Theoretically not justified
  - The compression scheme of STS and TTT may have different nature
- What does the attention score mean after the switch?

Proposed model- before the switch epoch

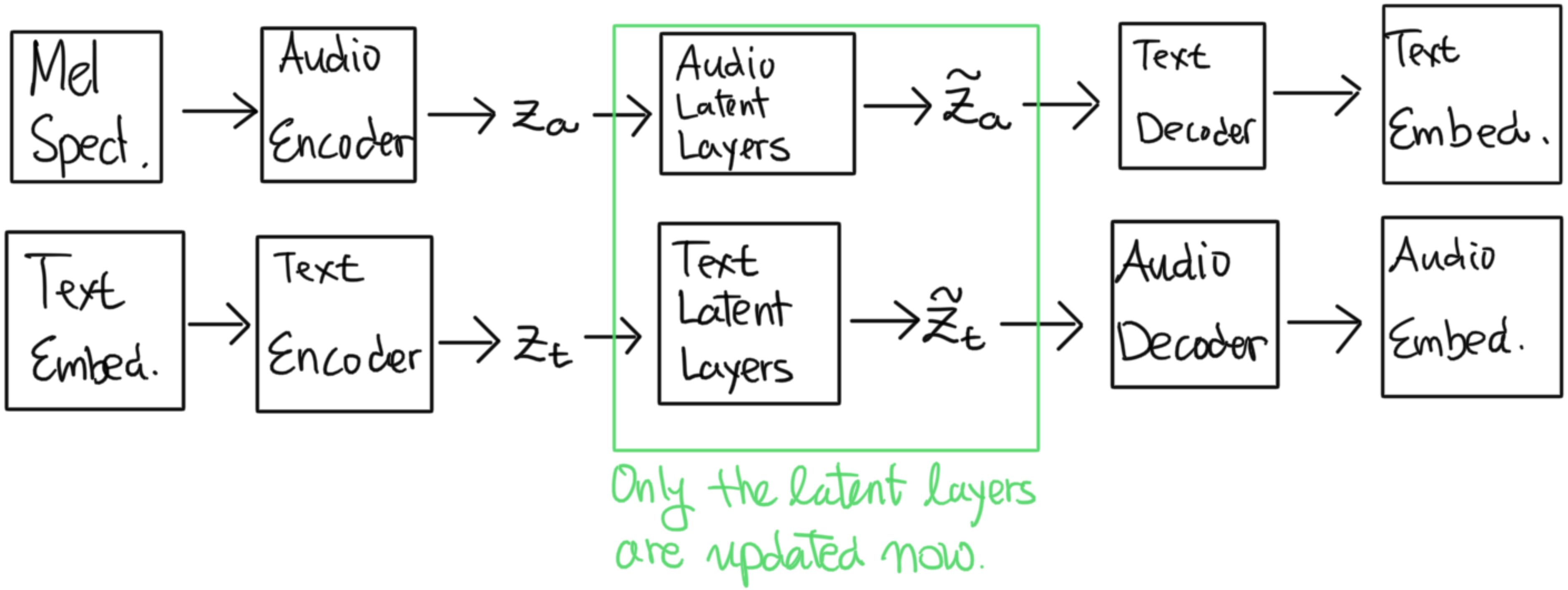


Loss functions?

- The model simply is two VAEs

- ELBO for each latent outputs z_a and z_t
- Cross-entropy for the text output
- ℓ_1 loss for the audio output
  - As done with the first Tacotron model

Proposed model – from the switch epoch



Parameter updates?

- Losses?
    - ELBO loss on $\tilde{z}_a$ and $\tilde{z}_t$
    - ℓ_1 loss on the audio output (i.e. the output resulting from $\tilde{z}_t$)
    - Cross-entropy on the text output (i.e. the output resulting from $\tilde{z}_a$)

- Update the parameters on the latent blocks only

My concerns

- Bothered 4 people already—hard payback if it doesn't work
- Can the transition from z to $\tilde{z}$ be Markovian?
- My goal is to make a model that is
  - Lego-like,
  - Light-weight,
  - Multi-speaker (by substituting different latent block?) and
  - Multilingual.

- Most importantly, Josh doesn't know about this yet…

Criticisms & Questions?

Thank you for listening